



AC_Branch_MobileNet: An effective hybrid branch network for plane classification in fetus brain ultrasound image

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ABSTRACT

Accurate classification of fetal brain planes in ultrasound imaging is essential for effective prenatal screening, enabling precise measurements and early detection of brain abnormalities. However, this task remains challenging due to dependence on sonographer expertise, variability in image quality, and complex anatomical structures. To address these issues, we propose the AC_Branch_MobileNet, a novel convolutional neural network incorporating a hybrid branch structure with an attention module, designed to automatically identify three key fetal brain planes: trans-ventricular, trans-cerebellar, and trans-thalamic. This method aims to improve performance in plane detection while reducing the parameter count to enhance system efficiency. Our approach first enhances image quality by reducing speckle noise and improving contrast using Maximum Local Variation-Based Unsharp Masking (MLVUM). MobileNetV2 is then employed as a lightweight baseline model for feature extraction, with an added branch structure to capture multi-level features without increasing model depth. In testing on a public fetal ultrasound dataset, our model achieved an accuracy of 95.11%, precision of 94.62%, recall of 94.39%, and F1-score of 94.50%. These results demonstrate that the proposed method outperforms the best baseline method by up to 7.43% in accuracy, 6.9% in precision, 6.72% in recall, and 6.81% in F1-score improvement. Overall, these findings indicate that AC_Branch_MobileNet effectively enhances accuracy while maintaining low parameter requirements, making it a promising tool for facilitating reliable fetal brain plane identification in clinical settings. Code is available at https://github.com/amenevatanparast/AC_Branch_MobileNet.

1. Introduction

The brain is a complex and vital organ that acts as the central control unit for human body functions, encompassing thoughts, emotions, movements, and physiological processes. Comprising billions of nerve cells known as neurons, the brain engages in intricate communication through electrical and chemical signaling pathways [1].

Throughout fetal development, the brain undergoes rapid growth and specialization, creating the complex array of structures that compose the central nervous system. Any abnormalities that occur during this crucial period can have profound effects on the development and function of the brain. Irregular fetal brain development can have significant short-term and long-term impacts on the newborn [2].

Fetal brain abnormalities can arise from genetic factors, environmental influences, or a combination of both. These abnormalities can manifest in various ways, such as structural malformations, disruptions in brain function, or neurodevelopmental disorders. Some common fetal

brain abnormalities include neural tube defects, genetic disorders like Down syndrome, and fetal alcohol syndrome [4]. These abnormalities can lead to a wide range of neurological and developmental impairments, including cognitive disabilities, motor deficits, and sensory disturbances. The impact of fetal brain abnormalities on an individual's life can vary greatly depending on the specific nature and severity of the condition.

Ultrasound (US) imaging technology has been used as a standard method for tracking the progress of the developing human fetus for almost 30 years, providing valuable insights into its development and general health [5]. During the various stages of pregnancy, ultrasound examinations are performed to confirm the pregnancy, assess the position, condition, dimensions, growth rate, alignment, and gestational age of the fetus, detect possible congenital disabilities and complications, and collect various other details that are necessary to ensure its health [6,7]. Among the prenatal examinations, fetal brain development is the most important. Fetal brain measurement at standard levels can directly

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reflect fetal growth [8] and fetal age [9]. Also, in the fetal brain development process, doctors can observe brain abnormalities using standard planes, such as ventricular expansion, intracranial hematomas, hydrocephalus, and subependymal cysts, which are usually manifestations of intracranial hemorrhage, ischemic diseases, and brain hypoxia [10]. Timely recognition of abnormal prenatal brain development holds significant implications for detecting fetal demise, and congenital anomalies, guiding appropriate diagnostic and therapeutic interventions, and even considering fetal termination [5], thereby reducing perinatal mortality and empowering parents to make informed decisions.

According to the guidelines of the International Society of Ultrasound in Obstetrics and Gynecology (ISUOG) [11], the use of standardized acquisition planes improves the reproducibility of (i) fetal biometry assessment and (ii) fetal evaluation. The planes commonly obtained for biometric extrapolation are the standard planes of the fetal abdomen (FASP), brain (FBSP), and femur (FFESP). FBSP includes standard trans-ventricular, trans-cerebellar, and trans-thalamic planes. Fig. 1 shows the standard planes of the brain. Note that these planes are the axial planes in imaging ultrasound.

The physiological evaluation of fetal brain abnormalities necessitates the expertise of trained sonographers capable of accurately preparing standard planes. Despite the development of a comprehensive quality control manual for standard design evaluation [12], the precision of measurements remains heavily reliant on the operator's skill and training experience [13]. This reliance highlights the pressing need for a medical support system dedicated to fetal plane diagnosis, as clinicians in clinical practice face considerable challenges in manually acquiring standard fetal planes. The complexity arises from the shared anatomical structures across various planes, which complicates the classification process, making the pursuit of a more efficient diagnostic system both crucial and advantageous.

While numerous articles have addressed the diagnosis of fetal planes [13,14,15], none have classically focused on feature extraction for classification purposes. Deep Learning (DL) has emerged as a robust tool for automatic diagnostics and medical image classification, leveraging its unique capabilities and advantages. Recently, deep learning algorithms have garnered significant attention for addressing various challenges in medical imaging [14,16], including fetal standard-plane classification [5,7,13,17], anatomical-structure analysis [18,19,20,21,22,23,24] and biometry parameter estimation [25,26,27,28,29].

Accurate diagnosis of fetal planes is vital in prenatal care, with deep learning methods significantly aiding their detection. Research has emphasized fetal plane classification, particularly the use of transfer learning in neural networks [5,7,15].

As illustrated in Fig. 2, several studies have employed deep learning techniques to classify fetal planes into six categories: abdomen, brain, femur, thorax, maternal cervix, and other (less frequently utilized planes). Additionally, several studies specifically address the classification of fetal brain planes, highlighting the diverse applications of deep learning methodologies in this critical area of prenatal diagnostics.

Recent advancements in fetal plane classification highlight the effectiveness of various deep-learning techniques aimed at improving diagnostic accuracy in ultrasound imaging. In their work, Krishna et al. [14] introduced a computer-aided diagnosis (CAD) algorithm that utilizes a stacked ensemble of deep convolutional neural networks (CNNs) for classifying standard fetal ultrasound planes. The methodology involves discrete CNNs for feature extraction, followed by predictions processed through SoftMax and a Random Forest classifier. This integrated approach yielded performance metrics, achieving an accuracy of 0.95, precision of 0.94, recall of 0.96, and an F1-score of 0.95. Kong et al. [30] implemented a dense network architecture to classify various fetal planes, including 4CH, FASP, FBSP, and FFESP. Their approach utilized a substantial dataset of 5,678 ultrasound images, achieving a precision of 0.98, recall of 0.97, and an F1-score of 0.98.

A multitasking framework, consisting of a faster R-CNN with an additional classification branch, is used to identify six key anatomical structures and assess head placement in the image [15]. This multitasking approach employed a faster R-CNN-inspired CNN to predict the presence of specific anatomical structures in abdominal, brain, and heart images. In this study, a dataset of 820 images was analyzed, resulting in a mean average precision (MAP) of 0.95 for structure recognition, with classification accuracy, precision, and recall values of 0.95, 0.95, and 0.93, respectively. In a different approach, Zhang et al. [13] explored multi-task learning by applying a Gaussian filter to smooth the original images before feeding them into a feature extraction network. The two networks then combine the information to produce the final result.

An integration of different deep learning algorithms is proposed by [31], where a committee of machines is formed using Gaussian Mixture Models (GMM) to group algorithms with similar performance levels. This committee then makes an overall classification decision through a Synergy Weighted Voting System (SWVS). The synergy of the

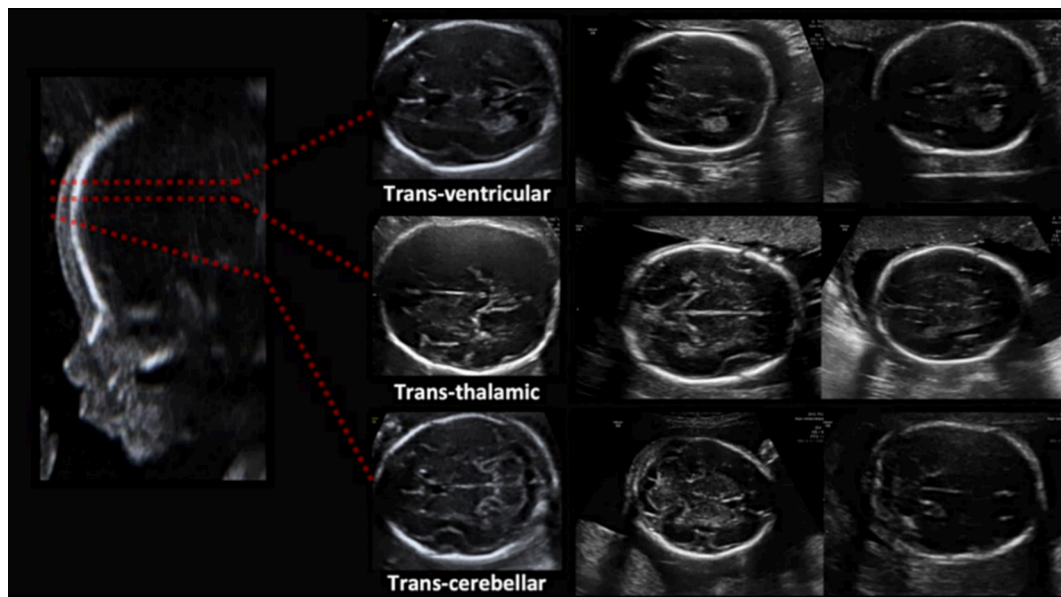


Fig. 1. The standard planes of the brain include trans-ventricular, trans-thalamic, and trans-cerebellar [3].

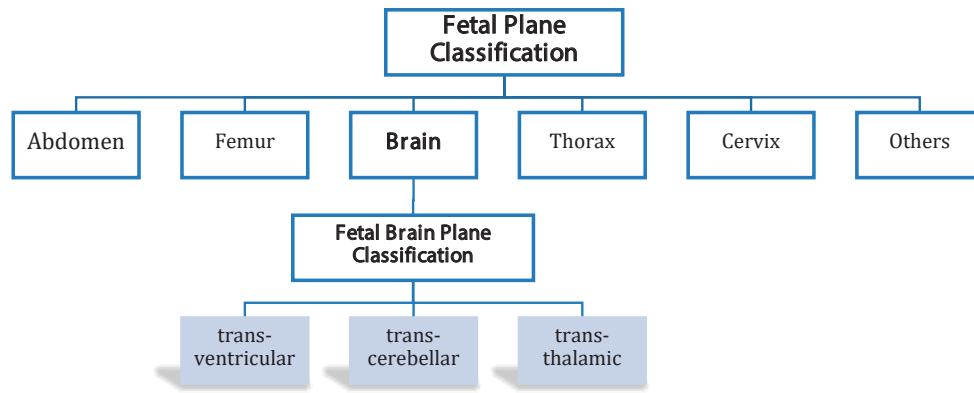


Fig. 2. Illustrates the general categorization of the reviewed articles. At the first level, deep learning techniques were employed to classify fetal planes into six categories: abdomen, brain, femur, thorax, maternal cervix, and others. At the second level, the focus transitioned to a more detailed classification of fetal brain planes, culminating in identifying three distinct groups: trans-ventricular, trans-thalamic, and trans-cerebellar.

framework involves the cooperation of selected deep learning algorithms in deterministic decision-making, based on their direct relationship with SWVS performance. They achieved an average accuracy of 0.90.

In addition to these methods, attention mechanisms have been utilized to help CNNs focus on the most distinctive regions in ultrasound fetal images, thereby improving standard plane detection performance and increasing the interpretability of deep learning results. The work [32] was among the first to incorporate a soft-attention mechanism into CNNs for identifying thirteen standard ultrasound planes, achieving an average accuracy of 0.98 across 38,243 images, along with precision, recall, and F1-scores of 0.93. Similarly, attention-guided CNNs, as proposed by Muna et al. [33], enhance the feature extraction processes in fetal plane classification.

Despite significant strides in classifying the standard fetal plane, challenges remain, particularly in accurately identifying distinct axial levels of the fetal brain and classifying brain images with abnormalities. These complexities stem from the intricate and overlapping structures within the fetal brain's axial planes, necessitating a more nuanced analysis compared to those of other fetal organ systems.

Fetal brain plane classification has seen significant advancements in recent years, particularly with the introduction of deep learning methodologies that have enhanced diagnostic accuracy and efficiency. This section reviews contributions in this field, categorizing studies by their methods.

Differential Convolutional Neural Networks: Ruowei et al. [5] proposed a differential CNN method to identify six standard and non-standard fetal brain planes. Unlike conventional CNNs that analyze images independently, their approach computes pixel-wise differences between two input images to construct a differential image for processing. Initially based on a dataset of 155 fetal images, the researchers employed data augmentation strategies to expand their dataset to 30,000 images, thereby mitigating concerns about overfitting. Their model achieved impressive performance metrics, with accuracy, precision, recall, and F1-score values of 0.93, 0.92, 0.92, and 0.93, respectively.

Generative Adversarial Networks (GANs): GANs have emerged as influential tools for improving fetal brain plane classification. In [34], the authors utilized a GAN in conjunction with a ResNet architecture to enhance fetal brain standard plane (FBSP) classification, achieving an accuracy of 0.81 and an F1-score of 0.80 upon validating their method with a dataset of 2,249 images. Building on this, the study presented in [35] employed a conditional GAN (cGAN) that utilized class activation maps (CAM) to generate fetal head standard planes (FHSP) artificially. By leveraging VGG16 to derive CAMs from FHSP images, the researchers achieved an average accuracy of $82.1 \pm 5.2\%$ for trans-ventricular, trans-thalamic, and trans-cerebellar plane classification. Additionally,

models designed for synthesizing fetal ultrasound brain plane images, such as diffusion-super-resolution GANs and transformer-based GANs, have been explored [36]. This innovative approach successfully generated 256x256 pixel trans-cerebellum brain images from ultrasound data, showcasing the versatility of GANs in enhancing image generation. To assist non-experts in locating and analyzing the trans-ventricular (TV) plane, [37] adopted a domain transfer learning approach that maps encoded spatial and temporal features from video sequences in the source domain to the desired standard plane representations in the target domain. This approach utilizes adversarial training to maintain image quality and incorporates a Denoising Probabilistic Diffusion Model followed by a Super-Resolution GAN, improving the clarity of generated outputs.

Automated Deep Learning Architectures: Automated deep learning architectures have been explored for fetal brain plane classification, aiming to enhance accuracy. In [7], an automated deep learning model was introduced for both fetal brain and normal maternal-fetal plane classification. This architecture integrates residual bottleneck blocks across 82 layers and employs Bayesian hyperparameter optimization to refine performance. Despite achieving classification accuracies of 0.78 and 0.79 for brain and normal fetal images, respectively, the model faces challenges related to computational efficiency, as its deep-layered structure results in significant processing time.

Segmentation-assisted deep learning has demonstrated an effective approach to fetal brain plane classification. In [38], a deep learning-based framework was proposed, using a U-Net model to segment the fetal brain region before classification. This segmentation step enhanced classification accuracy by reducing irrelevant features and improving feature extraction. The study utilized an ensemble model that combined Inception V3, ResNet50-V2, and DenseNet-201 with a max voting strategy. The segmentation step proved instrumental in refining the classifier's performance, allowing the model to focus on the relevant anatomical structures. The deep ensemble architecture, which combines Inception V3, ResNet50-V2, and DenseNet-201, increases computational costs, memory usage, and training time. While U-Net segmentation enhances accuracy, it also introduces preprocessing complexity and requires large annotated datasets for effective generalization.

Accurately recognizing fetal brain planes is vital for diagnosing abnormalities, as specific structures within each plane can indicate potential issues. High detection accuracy is essential; however, deeper networks in deep learning improve performance by increasing trainable parameters, which also leads to significant computational demands and lengthy training times. Despite these enhancements, many models still exhibit low accuracy and performance, limiting their practical utility. Therefore, optimizing model accuracy while simplifying network structures and reducing parameters is crucial for developing effective classification systems. Addressing these challenges is key to advancing

the reliability and applicability of fetal brain plane classification in clinical settings.

In this research, we aim to develop a network that effectively balances critical factors such as the number of parameters, memory consumption, and accuracy in the domain of fetal brain ultrasound analysis. To achieve this, we introduce AC_Branch_MobileNet, a novel hybrid branch convolutional neural network designed to optimize fetal brain plane classification by combining an attention module and a convolution module in parallel. While many deep learning approaches apply attention mechanisms sequentially, our design enhances feature extraction efficiency without increasing network depth. To further improve classification performance, we integrate Maximum Local Variation-Based Unsharp Masking (MLVUM) as a preprocessing step, significantly enhancing ultrasound image quality by reducing speckle noise and improving contrast—a crucial factor in medical image analysis. Rather than relying on computationally expensive feature extractors, our model employs MobileNetV2 solely as an encoder, leveraging its lightweight architecture for efficient deep feature extraction. However, the core novelty of our approach lies in the hybrid branch structure, where an attention module enhances salient feature representation while a convolution module captures essential spatial patterns, both operating in parallel to optimize learning without increasing model complexity. By combining these extracted features with global average pooling and dropout mechanisms, we successfully reduced the number of parameters in the final classification layers while maintaining high accuracy. Our method achieves a 7.43 % improvement in accuracy compared to the best baseline approach, demonstrating that AC_Branch_MobileNet significantly outperforms existing models in both classification accuracy and computational efficiency. These innovations improve our model's suitability for clinical applications requiring swift and accurate fetal brain plane identification. By optimizing computational efficiency and reducing the number of parameters, our approach ensures broad applicability across diverse systems, including those in underprivileged regions with limited technological infrastructure.

The remainder of the paper is organized as follows: Section 2 describes the proposed framework, including deep feature extraction, branch structure, and classification. Section 3 details the fetal ultrasound image dataset, presents the performance results of the proposed method evaluated on this dataset and compares these results with those of other competing methods. Discussions are presented in Section 4, and finally, the conclusion and future work is provided in Section 5.

2. Methodology

The primary goal of this paper is to introduce an architecture that decreases the need for an experienced sonographer, enabling the automatic classification of fetal brain planes into three distinct categories. By prioritizing adaptability and ease of use, this framework aims to expand the reach of fetal brain evaluations, ultimately contributing to better health outcomes for mothers and their babies.

To effectively classify ultrasound fetal brain images, our proposed method encompasses four steps: (i) image quality enhancement to ensure optimal clarity and detail, (ii) deep fetal ultrasound image feature extraction that is essential for accurate analysis, (iii) branch structure with an attention block that is crucial for enhancing the accuracy, (iv) a classification process to accurately categorize and interpret the data. A general view is shown in Fig. 3, and we will provide detailed explanations of each component. By small deep feature extraction and branch structure implementation, we have developed a fast network with a low parameter count that delivers impressive accuracy. This versatility allows seamless integration into various systems and devices, making it a valuable tool for healthcare professionals.

2.1. Image quality enhancement

As depicted in Fig. 4, the ultrasound images exhibit low quality due to speckle noise [39]. Speckle noise in ultrasound imaging appears as a granular texture, resulting from the constructive and destructive interference of backscattered echoes from organs that are often much smaller than the spatial resolution of a medical ultrasound system [40]. The low contrast levels further degrade the quality of the ultrasound images, complicating the analysis and interpretation of crucial anatomical features [41].

To address these challenges, this study employs Maximum Local Variation-Based Unsharp Masking (MLVUM) as a preprocessing step. Recent self-supervised image denoising methods, such as Blind Spot Networks and Transformer-based approaches, effectively reduce noise without the need for noisy-clean pairs [42]. However, in medical imaging, preserving diagnostic detail is critical. MLVUM ensures reliable noise reduction while maintaining essential anatomical features, thereby enhancing both the clarity and diagnostic utility of the ultrasound images.

MLVUM has the best performance in improving the quality of fetal brain ultrasound images [41]. By utilizing this method, we can significantly reduce noise in fetal brain ultrasound images, thereby enhancing both the overall quality and accuracy of our analyses.

The output Y of the MLVUM method is calculated by summing the input image X and the multiplication result of variable D and the scale λ that controls how the sharpening operation will affect the image [44].

$$Y = X + \lambda D \quad (1)$$

Variable D is generated by subtracting the output of the Maximum Local Variation-controlled Edge Smoothing Gaussian Filter (MLVESGF) from the input image X , as in Eq. (1). The normalized MLV $\nabla(m, n)$ is used to weigh it. MLV at (m, n) ($\nabla(m, n)$) is the highest absolute value of the differences between the intensity of the pixel at (m, n) and the intensity of its eight neighboring pixels.

$$D(m, n) = [X(m, n) - F(m, n)] \nabla(m, n) \quad (2)$$

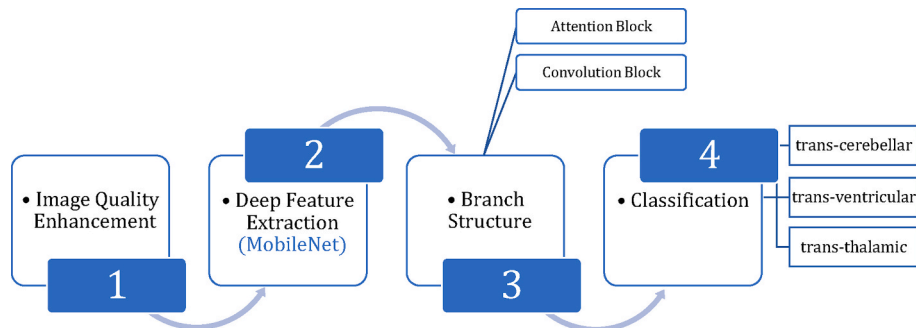


Fig. 3. A general view of AC_Branch_MobileNet. (1) image quality enhancement, (2) deep fetal US image feature extraction, (3) branch structure with an attention block, (4) a classification process to accurately categorize and interpret the data.

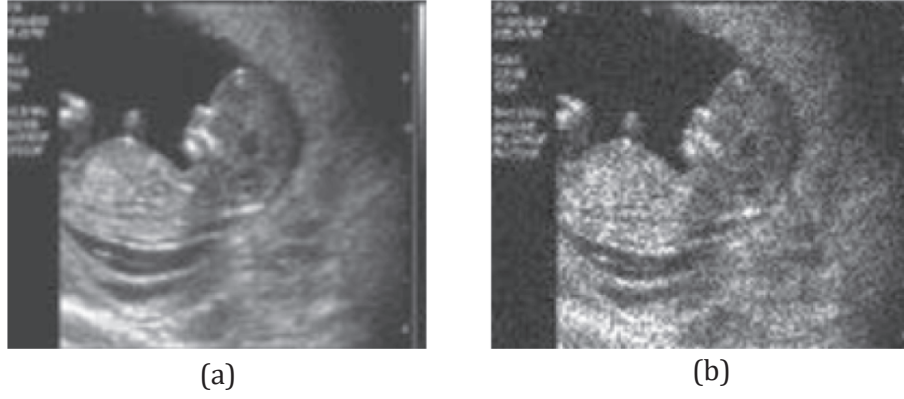


Fig. 4. The effect of speckle noise on image quality (a) Original image, (b) Image with speckle noise [43].

Where $m = 1, 2, \dots, M$ and $n = 1, 2, \dots, N$. M is the number of image rows, while N is the number of image columns. MLVESGF is a locally adaptive 2D Gaussian smoothing kernel with a standard deviation proportional to the normalized MLV local value. The output of the MLVESGF denoted as F is stated in Eq. (3).

$$F(m, n) = \sum_{i=-R}^{+R} \sum_{j=-R}^{+R} H_{mn}(i, j) X(m+i, n+j) \quad (3)$$

MLVESGF at a specific pixel location is stated in Eq. (4). R is the radius of MLVESGF, while σ_{mn} is the standard deviation of MLVESGF.

$$H_{mn}(i, j) = e^{-\left(\frac{i^2+j^2}{2(\sigma_{mn})^2}\right)} \Bigg/ \sum_{i=-R}^{+R} \sum_{j=-R}^{+R} e^{-\left(\frac{i^2+j^2}{2(\sigma_{mn})^2}\right)} \quad (4)$$

The results of applying the MLVUM method to the fetal brain US images are presented in Fig. 5. The image distinctly demonstrates enhanced edges, producing a sharper overall appearance, while effectively minimizing the presence of speckle noise.

2.2. Deep feature extraction

One significant challenge in computer vision tasks for medical image analysis is the collection of annotated image data, which is challenging and time-consuming. To address this challenge, we have taken two key steps. Firstly, we adopted a transfer learning approach to handle the need for a large amount of data in training deep convolutional neural networks (CNNs). Additionally, we opted for a network with fewer parameters, as the strength of a model is not solely determined by the maximum number of parameters it possesses [45].

For this purpose, we strategically selected MobileNetV2 as our baseline model for feature extraction after thoroughly evaluating various models. MobileNets are characterized by a streamlined architecture that employs depth-wise separable convolutions, enabling the construction of lightweight deep neural networks [46]. Among the models considered, MobileNetV2 stands out due to its low parameter count, totaling approximately 3.4 million parameters, making it an efficient choice for our objectives. This design choice not only enhances the model's efficiency but also aligns with our goal of optimizing performance while reducing computational overhead. By integrating MobileNetV2 into our framework, we aim to enhance the accuracy of our deep learning models for medical image analysis, ultimately achieving a harmonious balance between parameter quantity and performance quality.

2.3. Branch structure

By increasing the depth of the network, the accuracy of classification

and the number of extracted features also increase. However, with increasing depth, the number of trainable parameters rises, leading to a decrease in convergence speed. As a result, employing a small baseline model becomes essential to strike a balance between accuracy and efficiency while ensuring proper mapping of the original model.

The proposed architecture aims to achieve optimal accuracy with minimal parameters by incorporating a branch structure instead of a sequential one. This design choice reduces complexity and enhances feature extraction within each branch, ultimately increasing system execution speed and accuracy. By extracting unique features in each branch and aggregating them, the need to increase the depth of the network is mitigated. Moreover, the hybrid feature extraction architecture enables the application of features at varying levels with different coefficients, maximizing the model efficiency [45].

In one of the branches, we use the attention module. The attention mechanism enables networks to focus more on important features while suppressing unnecessary ones. This significantly improves the performance of networks. Not only does it tell where to focus, but it also increases the power of interest representation [47]. Table 1 shows the details of each layer of the attention module. As shown in this table, it is mainly composed of 1x1 convolutional layers with the exponential linear unit (ELU) activation. The number of filters of the conv layers is equal to 128, 32, and 1. ELU is a more complex activation function than ReLU. It returns the input if positive, otherwise returns an exponential function of the input minus one. The exponential function helps avoid the “dead neurons” problem that can occur with ReLU. It is shown that ELU performs better than ReLU in deep neural network architecture. It has a smooth curve that makes it easier to optimize compared to other activation functions [48]. ELU definition is given in Eq. (5). It includes an additional positive constant α , usually set to 1.0, which determines the saturation level of negative inputs.

$$f(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases} \quad (5)$$

In the other branch, we employ the convolution module to extract more features. The details of our architecture are outlined in Table 1. The convolution module largely consists of 3×3 convolutional layers followed by ELU activations. The number of filters in the convolutional layers is 64, 32, and 16, respectively. Fig. 6 illustrates the general architecture of the proposed deep model.

Our design leverages feature inputs from side branches and integrates them into a unified framework, creating a parallel network that facilitates multi-level feature applications. This approach enables the simultaneous extraction of diverse features from the side branches, resulting in a distinct set of features that varies according to the number of filters and iterations. The creation of various features based on different patterns has a more significant impact compared to utilizing a single wide network [45].

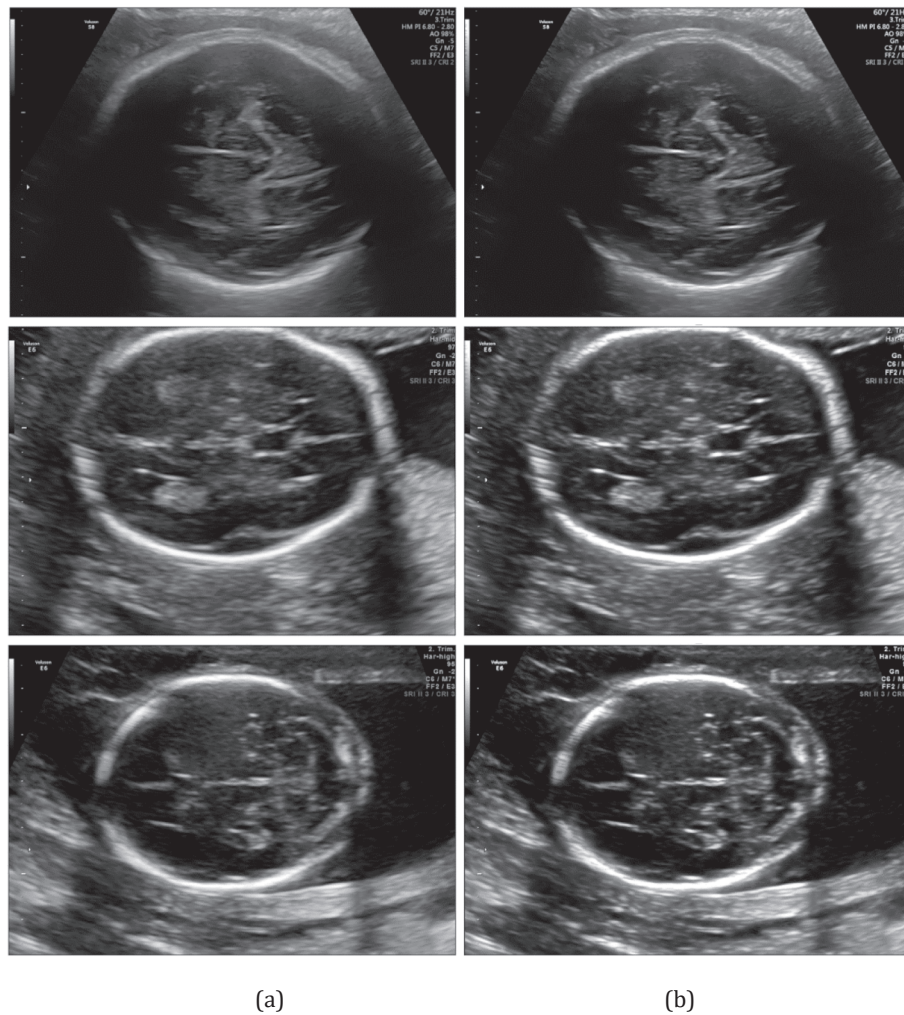


Fig. 5. Enhanced US images of the fetal brain (a) Input image, (b) Enhanced image with MLVUM.

Table 1

Layers of the AC Branch_MobileNet architecture. Conv, Pool, and Global_Ave_Pool refers to the Convolution operation, Pooling, and the Global Average Pooling Layers, respectively. Dense represents a fully connected layer.

Network parts	Layer (type)	kernels, activation	Output Shape	#Params
Pre-trained	Base model = MobileNetV2		(None, 8, 8, 1024)	3,360,451
Branch-structure	Attention-module	Conv1 $\times$ 1	(None, 8, 8, 128)	131,200
		Conv1 $\times$ 1	(None, 8, 8, 32)	4128
		Avg-Pool	(None, 8, 8, 32)	0
		Conv1 $\times$ 1	(None, 8, 8, 1)	33
	Convolution-module	Multiply	(None, 8, 8, 1024)	0
		Conv3 $\times$ 3	(None, 8, 8, 64)	589,888
		Conv3 $\times$ 3	(None, 8, 8, 32)	18,464
		Max-Pool	(None, 8, 8, 32)	0
		Conv3 $\times$ 3	(None, 8, 8, 16)	4624
Classification	concatenate	Attention, Convolution	(None, 8, 8, 1040)	0
	Global_Ave_Pool	--	(None, 1040)	0
	Dropout	--	(None, 1040)	0
	Dense	32, ELU	(None, 128)	133,248
	Dropout	--	(None, 128)	0
	Dense	3, Softmax	(None, 3)	387
Total Parameters		4,110,836		

2.4. Classification

After collecting the features, the Global Average Pooling (GAP) operation is utilized, which averages the output of each feature map

from the preceding layer by setting pool_size equal to the size of the input feature map. This approach eliminates the need for fully connected layers on top of the feature maps, opting instead to average each feature map. Unlike traditional down-sampling methods that focus on patches of

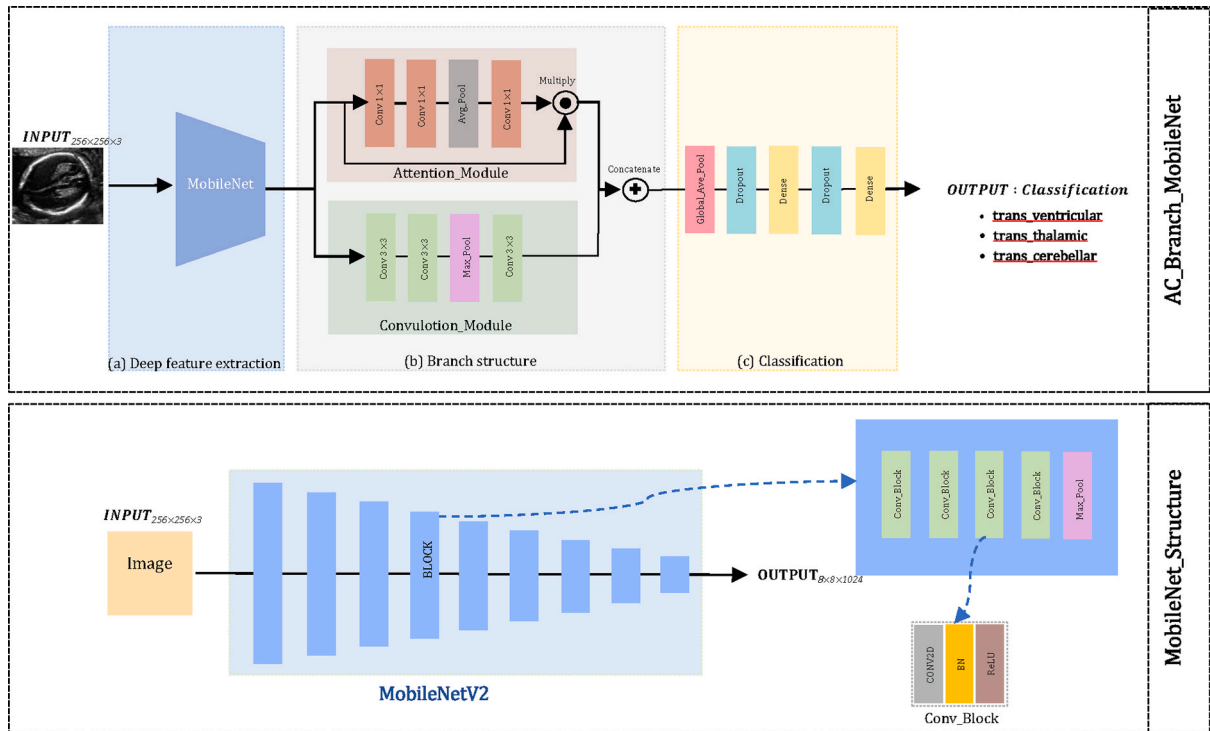


Fig. 6. The general architecture of the proposed ACBranch_MobileNet model: (a) Deep feature extraction with MobileNet. (b) The branch structure includes the attention module represented by the brown box and the convolution module defined by the green box. (c) During classification, features are aggregated, followed by Global Average Pooling (GAP), Dropout (DO), and Fully Connected (FC) layers. Finally, a fully connected layer with three neurons and Softmax activation is used to classify images as trans-ventricular, trans-thalamic, or trans-cerebellar.

the input feature map, global pooling down-samples the entire feature map to a single value. Following the GAP process, a dropout (DO) layer is introduced to mitigate overfitting in the neural network by randomly deactivating nodes during the training epochs.

Subsequently, a fully connected (FC) layer is integrated, where each neuron applies a linear transformation to the input vector using a weight matrix, establishing all possible layer-to-layer connections. This comprehensive connectivity ensures that every input of the input vector influences every output of the output vector. Another dropout layer is then implemented before culminating in a final fully connected layer containing three neurons with Softmax activation. This final layer is essential for classifying images into distinct categories, specifically identifying them as trans-ventricular, trans-thalamic, or trans-cerebellar planes.

3. Result analysis

This section outlines the experiments to validate the proposed method for classifying ultrasound fetal brain plane images. It begins with a comprehensive overview of the dataset utilized, detailing the characteristics and composition of the data. Subsequently, implementation settings and evaluation metrics to assess the efficiency and accuracy of the proposed methodology are articulated. The final subsection delves into the analysis of the results obtained from the experiments.

3.1. Dataset

Burgos-Artizzu et al. [3] compiled an extensive public dataset¹ of maternal and fetal screening ultrasound images, categorizing them into six classes. These classes include the four most prevalent fetal

anatomical planes (abdomen, brain, femur, and thorax), maternal cervix, and a general category. Within the fetal brain image class, further subdivisions were made into three common fetal brain planes (trans-thalamic, trans-cerebellum, trans-ventricular) to assess fine-grained classification performance. The dataset includes over 12,400 images; however, since our analysis focuses specifically on brain images, we select a subset of 2,950 brain images.

3.2. Data augmentation

Convolutional Neural Networks (CNNs) provide extensive advantages in extracting learned features from images; however, their deep structures and reliance on supervised learning can lead to overfitting, especially when working with limited training data—a challenge frequently encountered in medical contexts. This study implements traditional data augmentation techniques to address this challenge. Traditional data augmentation is preferred in medical imaging for its safety, efficiency, and control, as it preserves diagnostic details without risking distortions. Basic transformations, unlike GANs, avoid artifacts and work well with small datasets, maintaining data integrity essential for accurate clinical models [49]. These techniques involve (i) Rotation: Rotating ultrasound images by angles of 90, 180, and 270 degrees; (ii) Flipping: Vertically or horizontally flipping ultrasound scans; and (iii) Zoom: Allowing for zooming in or out (up to a maximum of 1.5 times its original dimensions) to maintain the overall brain structure integrity in the ultrasound scans.

In our research, our primary focus was on analyzing the fetal brain planes depicted in the acquired images. Specifically, these images were categorized into three prevalent fetal brain planes: trans-thalamic, trans-cerebellum, and trans-ventricular. The collective representation of these three planes encompassed a total of 2950 instances. Following the completion of all preprocessing tasks and data augmentation strategies applied to the initial dataset, we amassed a total of 4770 brain images distributed evenly across the three distinct classes.

¹ <https://zenodo.org/records/3904280>.

3.3. Training

Transfer learning was applied in our study using the network parameters of the pre-trained MobileNetV2 model, which was initially trained for image classification on the ImageNet dataset. Our goal was to assess its effectiveness in predicting fetal brain planes following training on our dataset. After initializing the network with pre-trained weights, it underwent an additional 50 epochs of training to strike a balance between achieving sufficient model convergence and preventing overfitting to our dataset. This process involved fine-tuning the parameters to enhance predictive accuracy for our specific task.

All models were developed utilizing TensorFlow v2.15, and the experiments were conducted on the Google Colab platform.

Categorical cross-entropy is a widely employed loss function in machine learning, particularly in classification tasks involving categorical output variables with multiple classes. This loss function quantifies the disparity between the actual probability distribution of classes and the predicted probability distribution. The mathematical formula for calculating categorical cross-entropy loss [50] is shown in Eq. (6).

$$\text{loss} = - \sum (y \times \log \hat{y}) \quad (6)$$

Where y is a one-hot encoded vector representing the true class labels, $\hat{y}$ is the predicted probability distribution over the classes, and the sum is taken over all classes [35].

Furthermore, we employed the Adaptive Moment Estimation (ADAM) optimization technique to enhance the optimization of network loss. Fig. 7 illustrates the training and validation progress of our AC_Branch_MobileNet model over the epochs. As shown in Fig. 7a, model accuracy consistently increases during training, accompanied by

a corresponding decrease in loss (Fig. 7b). The validation loss remains stable, reaching a low value (Min Val Loss: 0.18), which indicates robust learning and effective generalization. This strong performance is achieved through strategic data augmentation (applied after data splitting to ensure independent validation), the inclusion of dropout layers, the use of Global Average Pooling to optimize the parameter count, and the lightweight MobileNetV2 architecture.

3.4. Model evaluation metrics

Quantitative metrics used to evaluate the classification performance of the proposed technique are accuracy, precision, recall, and F1-score. Model accuracy is the ratio of the number of correctly classified samples to the total number of ground truth samples. Precision is the true positive rate, and recall is the measure of correctly identifying true positives. The harmonic mean of precision and recall scores is called the F1-score. The mathematical representation of the above evaluation criteria is defined in Eq. (7) to Eq. (10).

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

where TP, TN, FP, and FN represent the number of true positives, true

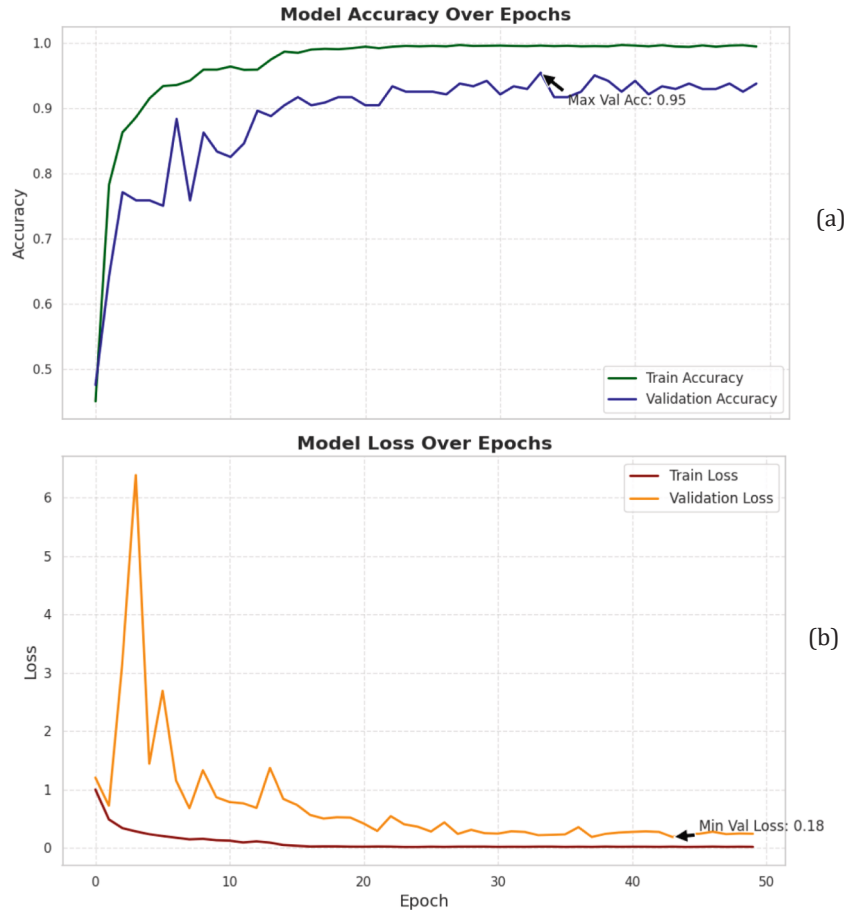


Fig. 7. (a) Accuracy vs. Epoch, and (b) Loss graph vs. Epoch.

negatives, false positives, and false negatives respectively.

3.5. Classification results

This article presents AC_Branch_MobileNet, a novel ultrasound brain image classification approach that utilizes a branch neural network for optimized depth. By aggregating features from two branches, this architecture reduces parameters while enhancing accuracy. The MobileNet framework balances computational efficiency with high accuracy, making it effective for datasets with limited samples.

The AC_Branch_MobileNet architecture features an innovative design that integrates an attention module with a branch structure. The attention module enhances feature extraction, while the branch structure combines an attention block and a convolution block that operate in parallel. This combination optimizes the network's depth and parameter count while improving classification accuracy. These features ensure efficient and effective performance across a variety of image-classification tasks.

To evaluate the impact of each component in our proposed AC_Branch_MobileNet architecture, we conducted an ablation study comparing several model configurations, as shown in Fig. 8. The models tested include MobileNetV2 (baseline), A-MobileNetV2 (MobileNetV2 + attention module), AC_Branch_MobileNetV2 (hybrid branch architecture), MobileNetV3_small, A-MobileNetV3_small, and AC_Branch_MobileNetV3_small.

MobileNetV2 served as the baseline model in our study, achieving an accuracy of 93.76 %. This model demonstrated strong feature extraction capabilities while maintaining a lightweight design, making it a solid foundation for further enhancements.

To improve feature representation, we introduced an attention mechanism, resulting in the A-MobileNetV2 model. This modification led to a noticeable performance increase, with accuracy improving to 94.20 %. This improvement highlights the ability of attention mechanisms to emphasize critical regions in ultrasound images, leading to more precise classification.

Building upon this, we developed AC_Branch_MobileNetV2 model, integrating both attention and convolution modules in a parallel hybrid structure. This approach achieved good performance among MobileNetV2-based models, with an accuracy of 95.11 %. The parallel hybrid architecture effectively balances spatial feature extraction (via convolution) and feature enhancement (via attention), leading to rich feature representations and improved classification performance. Additionally, global average pooling and dropout mechanisms played a crucial role in maintaining computational efficiency while reducing

overfitting, making this model suitable for real-time clinical applications.

While exploring more lightweight alternatives, we evaluated MobileNetV3_small, which significantly reduced the parameter count from approximately 4.4 million (MobileNetV2) to 1.1 million. However, this reduction resulted in a slight performance decrease, achieving 92.85 % accuracy. Introducing an attention mechanism in A-MobileNetV3_small improved this metric to 93.67 %, reinforcing the benefits of attention mechanisms. The AC_Branch_MobileNetV3_small model, which combines both attention and convolution modules, achieved 94.02 % accuracy.

Although this represents an improvement over the baseline MobileNetV3_small, it did not surpass the performance of AC_Branch_MobileNetV2. This outcome highlights that for the intricate task of distinguishing between the trans-ventricular, trans-thalamic, and trans-cerebellar planes, the slightly higher parameters count of AC_Branch_MobileNetV2 (approximately 4.1 million) provides the necessary capacity to extract more nuanced and discriminative features, leading to its superior diagnostic accuracy. Additionally, Fig. 9 illustrates the comparative performance of these architectures. These findings validate the effectiveness of our hybrid branch design and demonstrate that each component contributes meaningfully to performance gains, particularly in distinguishing visually similar fetal brain planes.

The trans-thalamic plane, due to its similarity to the trans-ventricular view and the presence of common structures with the trans-cerebellum plane, plays a central role in accurate classification. As illustrated in Fig. 9, the AC_Branch_MobileNetV2 model effectively reduces misclassifications of trans-ventricular samples as trans-thalamic (AC_Branch_MobileNetV2 misclassified 14 cases, which is fewer than the 15 and 25 cases misclassified by other MobileNetV2) and vice versa (AC_Branch_MobileNetV2 misclassified 11 cases, which is fewer than the 20 and 15 cases misclassified by other MobileNetV2). This underscores the model's proficiency in distinguishing between these visually similar views, enhancing overall classification accuracy, and enabling precise identification of fetal brain structures within ultrasound images.

The utilization of diverse datasets, some of which are of limited size, poses a challenge in conducting a fair comparison of the proposed solutions in the existing literature. To address this issue, the study opted to use identical datasets for a more robust comparison. Table 2 provides a performance evaluation of the number of parameters and classification accuracy in pre-trained deep convolutional neural network (CNN) models. This evaluation encompasses models such as the integration of ResNet with a Generative Adversarial Network (GAN) [34], VGG16 with conditional Generative Adversarial Network (cGAN) [35], an automated

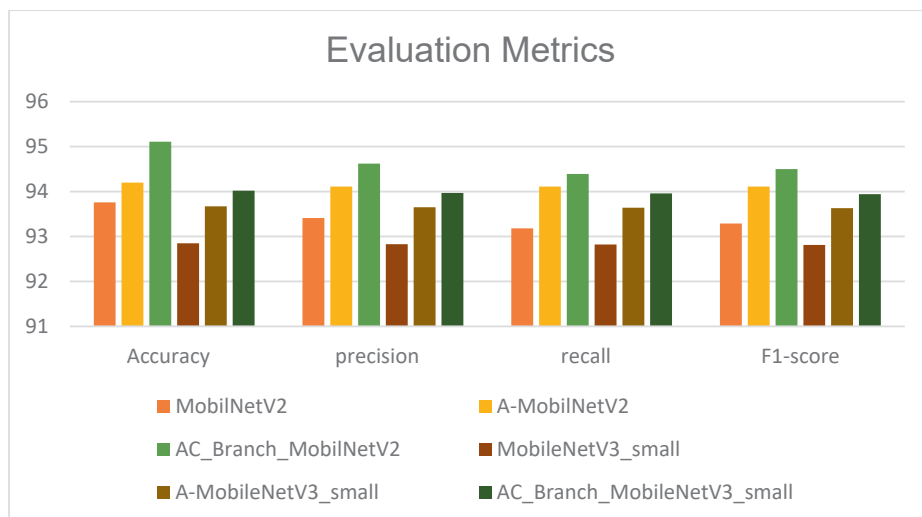


Fig. 8. Investigation of the impact of specific components within the AC_Branch_MobileNet architecture.

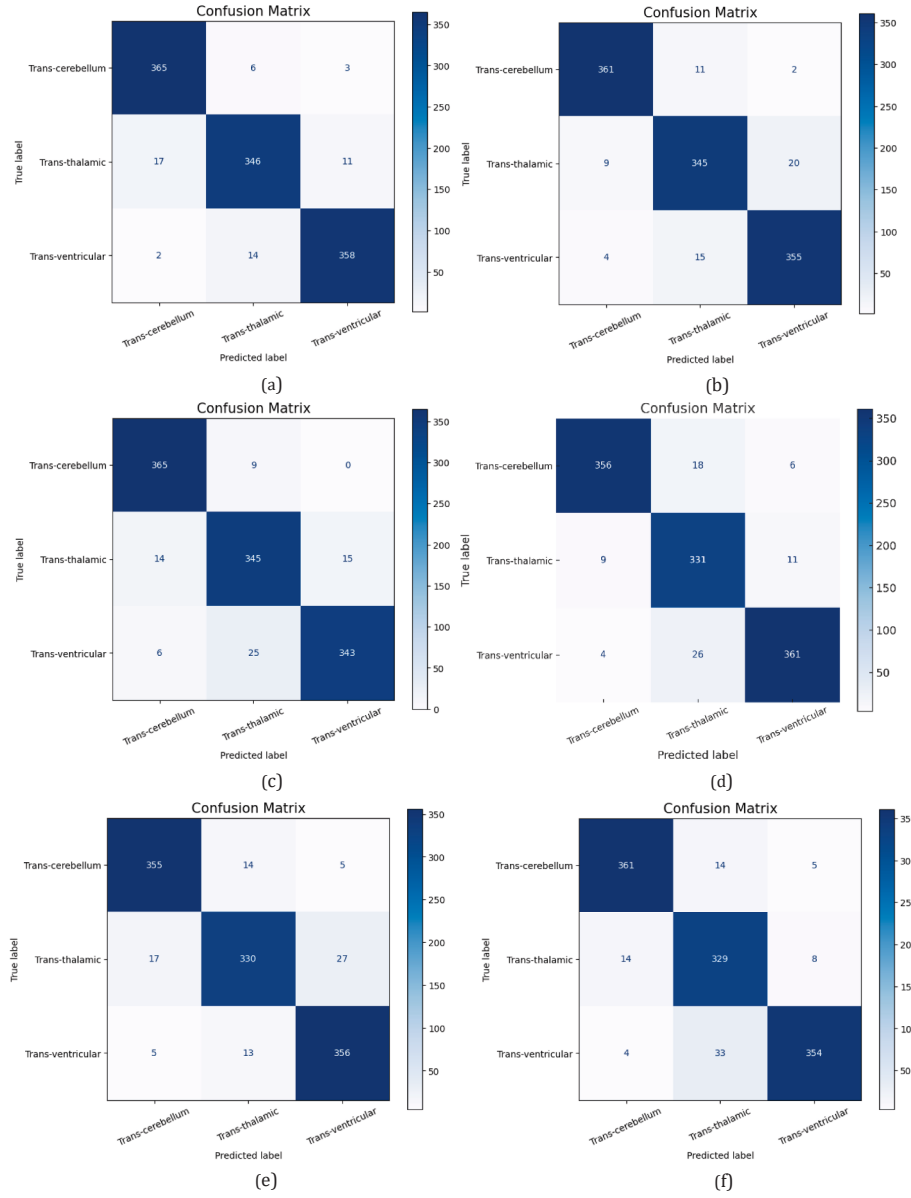


Fig. 9. Confusion matrices comparing the classification performance of six models: (a) AC.Branch.MobileNetV2, (b) A-MobileNetV2, (c) MobileNetV2, (d) AC.Branch.MobileNetV3_small, (e) A-MobileNetV3_small, and (f) MobileNetV3_small. These matrices illustrate the performance of each model in accurately distinguishing between the trans-ventricular, trans-thalamic, and trans-cerebellar fetal brain planes. Among them, the AC.Branch.MobileNetV2 model shows fewer misclassifications between the trans-thalamic and trans-ventricular planes.

Table 2

The comparison of the proposed architecture with several other state-of-the-art optimization algorithms has been carried out. The total number of parameters for the models in [7,34,38], and [35] has been estimated based on the layer configurations provided in those respective papers. The implementations of these advanced algorithms were conducted as described in the original studies, with the primary focus on applying them to our dataset for comparative analysis. This approach facilitates a fair evaluation of the proposed model against existing methods using consistent experimental conditions.

Papers	Planes			Accuracy	Metrics		F1-score	#Params
	TT	TV	TC		precision	recall		
[34]- 2021	✓	✓	✓	81.00	---	---	80.00	≈ 55M
[35]- 2023	✓	✓	✓	82.10	79.33	82.00	80.34	≈ 155M
[7]- 2023	✓	✓	✓	79.7	40.26	39.42	39.69	≈ 78M
[38]- 2024	✓	✓	✓	87.68	87.72	87.67	87.69	≈ 71M
MobileNetV3_small	✓	✓	✓	92.85	92.83	92.82	92.81	1,046,003
A-MobileNetV3_small	✓	✓	✓	93.67	93.64	93.64	93.63	1,062,580
AC_MobileNetV3_small	✓	✓	✓	94.02	93.97	93.96	93.96	1,159,460
MobileNetV2	✓	✓	✓	93.76	93.41	93.18	93.29	3,360,451
A-MobileNetV2	✓	✓	✓	94.20	94.11	94.11	94.11	4,107,062
AC_Branch_MobileNetV2 (Ours)	✓	✓	✓	95.11	94.62	94.39	94.50	4,110,836

deep learning bottleneck residual [7], and U-Net Ensemble Classification method [38].

The competing methods in Table 3 are fine-tuned on the same fetal brain ultrasound image dataset to eliminate dataset-related biases. To ensure a fair and meaningful comparison with the competing methods, we adhere to the optimal configurations and parameters (e.g., loss function, optimization algorithm, batch size, learning rate) and evaluation results as outlined in their original studies or executed with similar parameters and conditions on the same datasets.

By maintaining a rigorous and consistent evaluation framework, this comparative analysis effectively highlights the strengths and limitations of each model, providing deeper insights into the challenges of fetal brain ultrasound image classification. To further illustrate model performance, Fig. 10 presents the confusion matrices, offering a comprehensive visualization of classification accuracy and misclassifications across different fetal brain planes.

To verify the accuracy of the AC_Branch_MobileNet model, we utilize Grad-CAM [51] (Gradient-weighted Class Activation Mapping) to create heatmap visualizations that identify key areas of focus in the ultrasound images. These heatmaps are overlaid on the original images to highlight the regions that the model deemed most important for its classification decisions. Fig. 11 presents examples from the test dataset, demonstrating how the model consistently directs its attention to relevant anatomical structures associated with each fetal brain plane. The highlighted regions indicate that the model not only accurately detects the information needed for classification but also effectively differentiates key features across different brain planes, thereby enhancing its interpretability and reliability in clinical applications.

The AC_Branch_MobileNet model demonstrated superior performance in ultrasound brain image classification, achieving an accuracy of 95.11 %, precision of 94.62 %, recall of 94.39 %, and F1-score of 94.50 %. It outperforms other state-of-the-art methods while maintaining a significantly smaller parameter count. Its architecture, combined with interpretability through Grad-CAM visualizations, ensures reliability and efficiency in clinical applications, effectively addressing the complexities of fetal brain ultrasound classification.

4. Discussions

The complexity of the proposed AC_Branch_MobileNet approach was carefully managed to strike an optimal balance between accuracy and computational efficiency. By leveraging the lightweight MobileNetV2 architecture as the backbone, the model achieves a significant reduction in parameter count (approximately 4.1 million) compared to competing methods, such as VGG16-cGAN and ResNet-GAN, which utilize around 155 million and 55 million parameters, respectively. Furthermore, the integration of a branch structure with attention and convolutional modules enhances feature extraction without increasing network depth, improving performance while maintaining low computational overhead. Additionally, image enhancement using Maximum Local Variation-Based Unsharp Masking (MLVUM) effectively mitigates speckle noise and low contrast, which is crucial for improving the accuracy of ultrasound image classification in medical diagnostics.

The AC_Branch_MobileNet approach offers notable benefits, including efficient feature extraction through MobileNetV2's streamlined design, which reduces both parameter count and computational demands. The results of AC_Branch_MobileNet, presented in Table 2, show an accuracy of 95.11 %, outperforming existing models such as the ResNet-GAN approach and automated deep learning methods with considerably higher parameter counts.

Notably, the model's hybrid architecture, which combines attention and convolutional modules, enhances feature recognition by emphasizing relevant details while suppressing less critical information. The robust performance across multiple evaluation metrics, including precision, recall, and F1-score at 94.62 %, 94.39 %, and 94.50 %, respectively, demonstrate the proposed method's capability to reliably support clinical workflows.

The confusion matrix (Fig. 9) illustrates the model's effectiveness in distinguishing between trans-ventricular, trans-thalamic, and trans-cerebellar planes. Notably, the model shows a reduced number of misclassifications when differentiating trans-thalamic from trans-ventricular planes, which are often challenging due to overlapping anatomical structures. This improvement suggests that the model's hybrid architecture, which combines attention and convolutional modules, enhances feature recognition by emphasizing relevant details while suppressing less critical information.

The Grad-CAM heatmap visualizations (Fig. 11) provide further insight into the model's decision-making process by highlighting the areas of the ultrasound images that contributed most to its predictions. The heatmaps indicate that the model focuses on critical anatomical regions associated with each plane, ensuring accurate identification. This not only confirms the model's effectiveness in capturing the essential features required for classification but also enhances its interpretability, a crucial factor in clinical applications. The robust performance across multiple evaluation metrics, including precision, recall, and F1-score by 94.62 %, 94.39 %, and 94.50 % respectively, demonstrate the proposed method's capability to reliably support clinical workflows.

The AC_Branch_MobileNet model offers significant clinical advantages by automating fetal brain plane classification with high accuracy and low computational demands. This efficiency reduces the reliance on expert sonographers, shortens examination times, and minimizes error rates. Its lightweight design makes it well-suited for portable ultrasound devices, allowing for broader access in remote or underserved areas. By enhancing diagnostic speed and reliability, AC_Branch_MobileNet supports more streamlined workflows and improved prenatal care.

To address the challenge of data scarcity, several advanced techniques have been proposed [52], including Transfer Learning (TL), Self-Supervised Learning (SSL), Generative Adversarial Networks (GANs), Physics-Informed Neural Networks (PINN), and the Deep Synthetic Minority Oversampling Technique (DeepSMOTE). Among them, this study adopts Transfer Learning (TL) due to its ability to significantly reduce the need for large labeled datasets, enabling effective training on domain-specific problems like fetal brain ultrasound image classification. Additionally, this study implements traditional data augmentation techniques, such as image rotation and flipping, to enhance the dataset.

Table 3
Hyperparameters of competing methods.

Method	Hyperparameters				Parameters #Params
	Loss function	Optimization Algorithm	Learning Rate	Batch Size	
[7]- 2023	categorical cross-entropy	Adam	0.001	128	≈ 78M
[34]- 2021	cross-entropy	Adam	0.01	64	≈ 55M
(GAN + ResNet-50)					
[35]- 2023	Weighted cross-entropy	Adam	0.001	32	≈ 155M
(VGG16 + Pix2Pix GAN)					
[38]- 2024	categorical cross-entropy	Adam	0.0001	64	≈ 71 M
(U-Net Ensemble)					
AC_Branch_MobileNet	categorical cross-entropy	Adam	0.001	32	≈ 4.1M

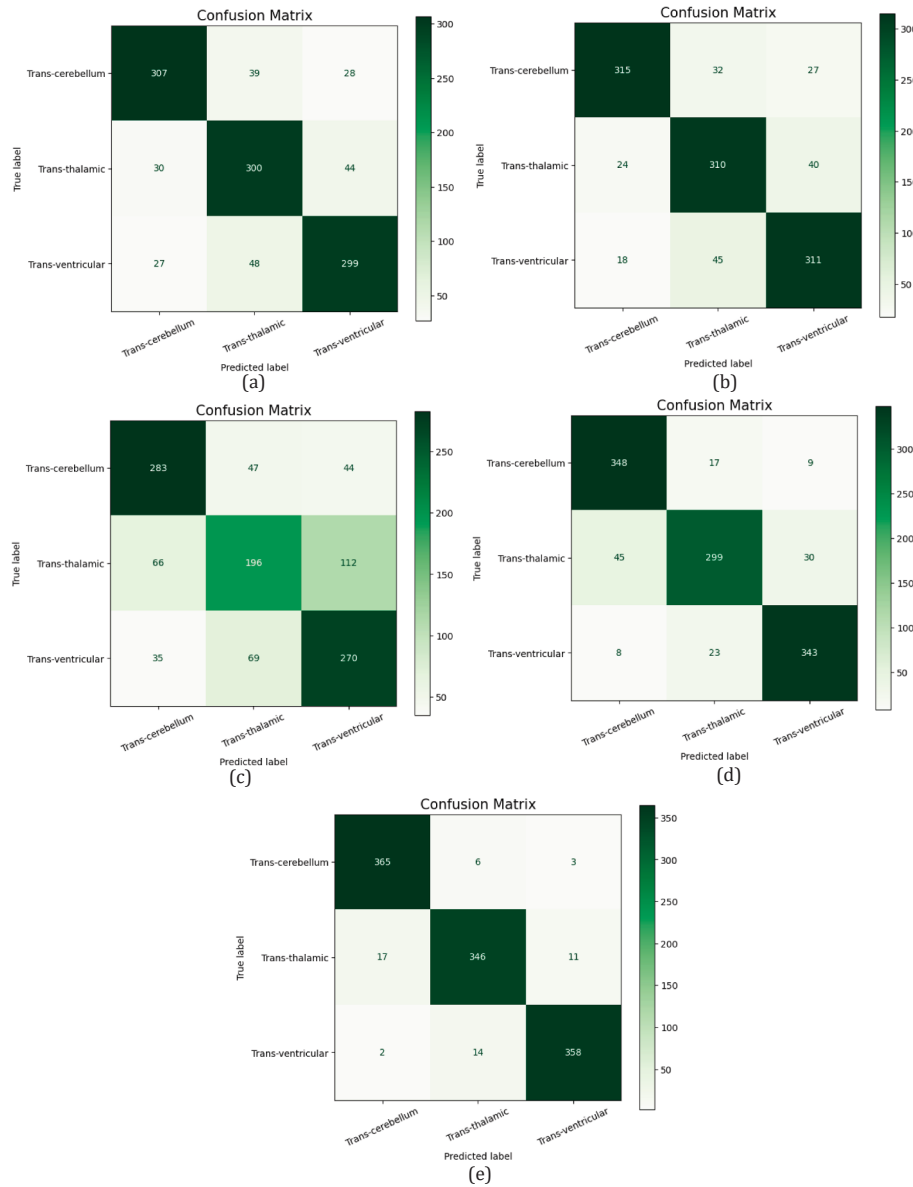


Fig. 10. Confusion matrices comparing the classification performance of different methods: (a) [34], (b) [35], (c) [7], (d) [38], and (e) AC_Branch_MobileNet (Ours).

Traditional augmentation is particularly suitable for medical imaging because it preserves diagnostic details and maintains label integrity, which is critical for clinical accuracy. Given these priorities, more complex generative approaches, such as GANs, were avoided due to the potential risks of introducing distortions that might compromise the reliability of the generated data.

While three-dimensional and four-dimensional ultrasound technologies have gained popularity, they are primarily used to provide expectant parents with visual images rather than for diagnostic purposes. Two-dimensional ultrasound remains the most authoritative standard for assessing fetal development, and AC_Branch_MobileNet directly enhances accuracy in classifying these critical two-dimensional ultrasound brain images [53].

5. Conclusion and future work

This paper presented an automated method for classifying fetal brain planes US images into three classes: trans-ventricular, trans-thalamic, and trans-cerebellum planes. The proposed approach employs an enhancement method for ultrasound images to reduce Speckle noise and

low contrast levels with Maximum Local Variation-based Unsharp Masking. Using MobileNet for feature extraction along with a branch structure allows features to be applied at different levels without increasing the depth of the network. Our goal was to introduce a network with a low number of parameters yet competitive accuracy through innovative methods, making it versatile and suitable for integration into various systems and devices. The proposed method exhibited enhanced performance metrics, including accuracy, precision, recall, and F1-score, achieving outstanding values of 95.11 %, 94.62 %, 94.39 %, and 94.50 %, respectively. Interestingly, the proposed framework can help sonographers, especially novices, enhance the efficiency of fetal brain plane ultrasound detection during prenatal screening. Future work will extend AC_Branch_MobileNet to segment brain structures using meta-learning, enabling the model to better adapt to anatomical variations and diverse datasets. The focus will be on segmenting key structures such as the ventricles and cerebellum, which will facilitate the early detection of neurological conditions. Additionally, integrating disease detection capabilities into the framework could enhance the identification of abnormalities, including Dandy-Walker malformation. The goal is to create a comprehensive tool that

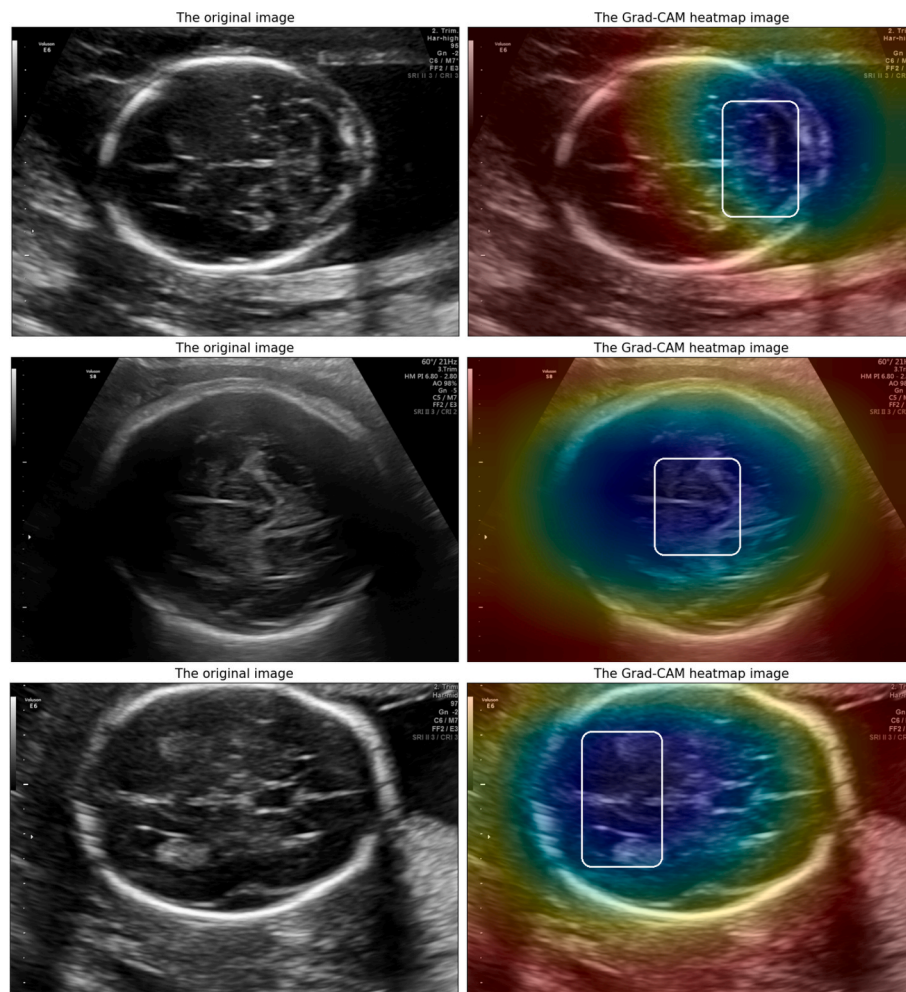


Fig. 11. Visual representation of the AC Branch MobileNet model's performance using heatmaps. (a) Original ultrasound image, and (b) Grad-CAM heatmap overlay, where blue regions indicate areas of high importance for the model's classification. The white box outlines key anatomical structures within each plane that contributed significantly to the decision-making process.

combines plane classification, segmentation, and disease detection for prenatal brain assessment.

CRediT authorship contribution statement

Amene Vatanparast: Resources, Formal analysis, Software, Conceptualization, Visualization, Methodology, Writing – original draft. **Mansoor Fateh:** Project administration, Writing – review & editing, Conceptualization, Validation, Supervision, Investigation. **Hoda Mashayekhi:** Writing – review & editing, Resources, Visualization, Investigation, Data curation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] W. Walz, *The Brain as an Organ*, Springer, *The Gliocentric Brain*, 2023, pp. 7–27.

- [2] H. Chen, P. Bai, S. Yang, M. Jia, H. Tian, J. Zou, X. Xiao, Short-Term and Long-Term Outcomes of Fetal Ventriculomegaly beyond Gestational 37 Weeks: a Retrospective Cohort Study, *J. Clin. Med.* 12 (3) (2023) 1065.
- [3] X.P. Burgos-Artizzu, D. Coronado-Gutiérrez, B. Valenzuela, E. Bonet-Carne, E. Eixarch, F. Crispí, E. Gratacós, Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes, *Sci. Rep.* 10 (2020) 10200.
- [4] L. Avagliano, V. Massa, T.M. George, S. Qureshy, G. Bulfamante, R.H. Finnell, Overview on neural tube defects: from development to physical characteristics, *Birth Defects Res.* 111 (10) (2019) 1455–1467.
- [5] R. Qu, G. Xu, C. Ding, W. Jia, M. Sun, Standard Plane Identification in Fetal Brain Ultrasound Scans using a Differential Convolutional Neural Network, *IEEE Access* 8 (2020) 83821–83830.
- [6] B.S. Prabakaran, P. Hamelmann, E. Ostrowski, M. Shafique, FPUS23: an Ultrasound Fetus Phantom Dataset with Deep Neural Network evaluations for Fetus Orientations, Fetal Planes, and Anatomical Features, *IEEE Access* 11 (2023) 58308–58317.
- [7] F. Rauf, M.A. Khan, A.K. Bashir, K. Jabeen, A. Hamza, A. Alzahrani, N. Alalwan, A. Masood, Automated deep bottleneck residual 82-layered architecture with Bayesian optimization for the classification of brain and common maternal fetal ultrasound planes, *Front. Med.* 10 (2023).
- [8] L. Zhang, S. Chen, C.T. Chin, T. Wang, S. Li, Intelligent scanning: automated standard plane selection and biometric measurement of early gestational sac in routine ultrasound examination, *Med. Phys.* 39 (8) (2012) 5015–5027.
- [9] A.I.L. Namburete, R.V. Stebbing, B. Kemp, M. Yaqub, A.T. Papageorgiou, J. A. Noble, Learning-based prediction of gestational age from ultrasound images of the fetal brain, *Med. Image Anal.* 21 (1) (2015) 72–86.
- [10] B. Putbrese, A. Kennedy, Findings and differential diagnosis of fetal intracranial haemorrhage and fetal ischaemic brain injury: what is the role of fetal MRI? *Br. J. Radiol.* 90 (1070) (2017) 20160253.
- [11] L.J. Salomon, Z. Alfrevic, F.D.S. Costa, R.L. Deter, F. Figueras, T. Ghi, P. Glanc, A. Khalil, W. Lee, R. Napolitano, A. Papageorgiou, A. Sotiradis, J. Stirnemann,

- A. Toi, G. Yeo, ISUOG Practice guidelines: ultrasound assessment of fetal biometry and growth, *Ultrasound Obstet. Gynecol.* 53 (6) (2019) 715–723.
- [12] S.L. Murphy, J. Xu, K.D. Kochanek, E. Arias, Mortality in the United States, *NCHS Data Brief* 328 (2018) 1–8.
- [13] B. Zhang, H. Liu, H. Luo, K. Li, Automatic quality assessment for 2D fetal sonographic standard plane based on multitask learning, *Medicine (Baltimore)* 100 (4) (2021) e24427.
- [14] T.B. Krishna, P. Kokil, Standard fetal ultrasound plane classification based on stacked ensemble of deep learning models, *Expert Syst. Appl.* 238 (2024) 122153.
- [15] Z. Lin, S. Li, D. Ni, Y. Liao, H. Wen, J. Du, S. Chen, T. Wang, B. Lei, Multi-task learning for quality assessment of fetal head ultrasound images, *Med. Image Anal.* 58 (2019) 101548.
- [16] Y. Jalali, M. Fateh, M. Rezvani, V. Abolghasemi, H. Anisi, ResBCDU-Net: a deep learning framework for lung CT image segmentation, *Sensors* 21 (1) (2021) 268.
- [17] T.B. Krishna, P. Kokil, Automated classification of common maternal fetal ultrasound planes using multi-layer perceptron with deep feature integration, *Biomed. Signal Process. Control* 86 (2023) 105283.
- [18] E. Jain, P. Kaushik, V. Kukreja, A.D. Sakshi, B. Goyal, “Fetal Diagnostics using Vision Transformer for Enhanced Health and Severity Prediction in Ultrasound Imaging,”, *Curr. Med. Imaging* (2025).
- [19] L. Chen, P. Bentley, K. Mori, K. Misawa, M. Fujiwara, D. Rueckert, Self-supervised learning for medical image analysis using image context restoration, *Med. Image Anal.* 58 (2019) 101539.
- [20] J. Tan, A. Au, Q. Meng and B. Kainz, “Semi-supervised Learning of Fetal Anatomy from Ultrasound,” in *Domain Adaptation and Representation Transfer and Medical Image Learning with Less Labels and Imperfect Data. DART MIL3ID*, Cham, 2019.
- [21] H. Dou, X. Yang, J. Qian, W. Xue, H. Qin, X. Wang, L. Yu, S. Wang, Y. Xiong, P.-A. Heng and D. Ni, “Agent with warm start and active termination for plane localization in 3D ultrasound,” in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2019*, 2019.
- [22] X. Shu, F. Chang, X. Zhang, C. Shao, X. Yang, ECAU-Net: Efficient channel attention U-Net for fetal ultrasound cerebellum segmentation, *Biomed. Signal Process. Control* 75 (2022) 103528.
- [23] R.V. Lalitha, K.P. Ponnekanti, R.R. Tetali, K. Kavitha, S. Rayudu, B.R. Kiran, Efficient adaptive enhanced adaboost based detection of spinal abnormalities by machine learning approaches, *Biomed. Signal Process. Control* 80 (2) (2023) 104367.
- [24] D. Vetrivel, R. Thenmozhi, Prenatal Diagnostics using Deep Learning: a dual Approach to Plane Localization and Cerebellum Segmentation in Ultrasound Images, *J. Clin. Ultrasound* 53 (4) (2025) 721–735.
- [25] J. Liu, Y. Tan, C. Wang, K. Li, G. Tan and C. Liu, “FastCRL: A Fast Network With Adaptive Fourier Transform and Offset Prediction for Fetal Crown-Rump Length Measurement and Position Estimation in Ultrasound Images,” *IEEE J Biomed Health Inform.* 2025.
- [26] M. Sinclair, C. F. Baumgartner, J. Matthew, W. Bai, J. Cerrolaza, Y. Li, S. Smith, C. Knight, B. Kainz, J. Hajnal, A. P. King and D. Rueckert, “Human-level Performance On Automatic Head Biometrics In Fetal Ultrasound Using Fully Convolutional Neural Networks,” in *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Honolulu, 2018.
- [27] R. Yibiao, D. Xiang, W. Zhu, F. Shi, E. Gao, Z. Fan, X. Chen, Deriving external forces via convolutional neural networks for biomedical image segmentation, *Biomed. Opt. Express* 10 (8) (2019) 3800–3814.
- [28] E.L. Skeika, M.R. Da Luz, B.J.T. Fernandes, H.V. Siqueira, M. Sguario, Convolutional Neural Network to Detect and measure Fetal Skull Circumference in Ultrasound Imaging, *IEEE Access* 8 (2020) 191519–191529.
- [29] J. Wang, Z. Fang, S. Yao, Y. Feng, Ellipse guided multi-task network for fetal head circumference measurement, *Biomed. Signal Process. Control* 82 (2023) 104535.
- [30] P. Kong, D. Ni, S. Chen, S. Li, T. Wang and B. Lei, “Automatic and Efficient Standard Plane Recognition in Fetal Ultrasound Images via Multi-scale Dense Networks,” in *Data Driven Treatment Response Assessment and Preterm, Perinatal, and Paediatric Image Analysis. PIPPI DATRA 2018*, Melbourne, 2018.
- [31] S. Belciug, D.G. Iliescu, Deep learning and Gaussian Mixture Modelling clustering mix. a new approach for fetal morphology view plane differentiation, *J. Biomed. Inform.* 143 (2023) 104402.
- [32] J. Schlemper, O. Oktay, M. Schaap, M. Heinrich, B. Kainz, B. Glocker, D. Rueckert, Attention gated networks: Learning to leverage salient regions in medical images, *Med. Image Anal.* 53 (2019) 197–207.
- [33] M. S. Al-Razgan, Y. A. Ali and E. M. Awwad, “Enhancing Fetal Medical Image Analysis through Attention-guided Convolution: A Comparative Study with Established Models,” *Journal of Dental Research*, vol. 3, no. 2, 2024.
- [34] A. Montero, E. Bonet-Carne, X.P. Burgos-Artizzu, Generative Adversarial Networks to Improve Fetal Brain Fine-Grained Plane Classification, *Sensors (basel)* 21 (23) (2021) 7975.
- [35] A. Lasala, M.C. Fiorentino, S. Micera, A. Bandini, S. Moccia, Exploiting class activation mappings as prior to generate fetal brain ultrasound images with GANs. In *Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2023.
- [36] M. Iskandar, H. Mannering, Z. Sun, J. Matthew, H. Kerdegari, L. Peralta and M. Xochicale, “Towards Realistic Ultrasound Fetal Brain Imaging Synthesis,” *ArXiv*, 2023.
- [37] Q. Men, H. Zhao, L. Drukker, A.T. Papageorgiou, J.A. Noble, “Towards Standard Plane Prediction of Fetal Head Ultrasound with Domain Adaption,” in *IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, Cartagena, Colombia, 2023.
- [38] M.N. Hasan, A.A. Hossain, Fetal Brain Planes Classification using Deep Ensemble transfer Learning from U-Net Segmented Fetal Neurosonography Images, *Int. J. Image, Graphics and Signal Processing(IJIGSP)* 16 (4) (2024) 74–86.
- [39] H. Al-Ghaib, R. Adhami, “On the digital image additive white Gaussian noise estimation,” in *International Conference on Industrial Automation, Information and Communications Technology*, Bali, Indonesia, 2014.
- [40] A.K. Bedi, R.K. Sunkaria, D. Mittal, Despeckling of ultrasound medical images using nonlinear adaptive anisotropic diffusion in nonsubsampling shearlet domain, *Biomed. Signal Process. Control* 14 (2014) 55–65.
- [41] R. F. Gusa, R. Hidayat and H. A. Nugroho, “Performance Analysis of Enhancement Methods on Fetal Ultrasound Images,” in *2023 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT)*, BALI, Indonesia, 2023.
- [42] D. Zhang, F. Zhou, F. Albu, Y. Wei, X. Yang, Y. Gu, Q. Li, Unleashing the Power of Self-Supervised image Denoising, A Comprehensive Review, *arXiv* (2023).
- [43] K. Karthikeyan, C. Chandrasekar, Speckle Noise Reduction of Medical Ultrasound Images using Bayesshrink Wavelet Threshold, *Int. J. Comput. Appl.* 22 (2011) 8–14.
- [44] D.R. Edla, V.R. Simi, J. Joseph, A Noise-robust and Overshoot-free Alternative to Unsharp Masking for Enhancing the Acuity of MR Images, *J. Digit. Imaging* 35 (4) (2022) 1041–1060.
- [45] E. Parcham, M. Fateh, HybridBranchNet: a novel structure for branch hybrid convolutional neural networks architecture, *Neural Network* 165 (2023) 77–93.
- [46] A.G. Howard M. Zhu B. Chen D. Kalenichenko W. Wang T. Weyand M. Andreetto H. Adam, “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications,” *ArXiv*, vol. abs/1704.04861, 2017.
- [47] K. Simonyan and A. Zisserman, “Very Deep Convolutional Networks for Large-Scale Image Recognition,” in *3rd International Conference on Learning Representations (ICLR 2015)*, *Computational and Biological Learning Society*, 2015.
- [48] Y. Bai, RELU-Function and Derived Function Review, *SHS Web of Conferences* (2022).
- [49] C. Shorten, T. Khoshgoufar, A survey on image Data Augmentation for Deep Learning, *J. Big Data* 60 (2019).
- [50] M. Kerkhof, L. Wu, P. Guilherme, S. Picek, No (good) loss no gain: systematic evaluation of loss functions in deep learning-based side-channel analysis, *J. Cryptogr. Eng.* 13 (2023) 311–324.
- [51] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, “Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,” in *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017.
- [52] L.S. Alzubaidi, J. Bai, A. Al-Sabaawi, J. Santamaria, A survey on deep learning tools dealing with data scarcity: definitions, challenges, solutions, tips, and applications, *J. Big Data* 10 (46) (2023) 4.
- [53] American Institute of Ultrasound in Medicine, AIUM practice guideline for the performance of obstetric ultrasound examinations, *J. Ultrasound Med.* 32 (6) (2013) 1083–1101.